MEMORANDUM

TO: Leadership **FROM:** Shridhar Pathak **DATE:** October 13, 2026 **RE:** Proposal: a starting evaluation methodology for document AI quality

Context

We are scaling document ingestion toward tens of thousands of documents, parsed by machine and served through AI question-answering. At that scale, nobody can read the output to check it. Quality becomes something we measure, or something we simply hope for.

The problem

A document pipeline can look healthy while being wrong in ways that matter: a table parsed with shifted columns puts the wrong number in front of a decision-maker; a Q&A system that never says "I don't know" will hallucinate confidently instead. These failures are silent. Without an evaluation methodology, we find out from users, not from our systems.

The proposal

Adopt a four-layer evaluation methodology as our starting standard for any document AI work:

1. **Parse fidelity.** Does the parsed output faithfully represent the

source? Measured per document: text coverage against the source, table structure accuracy (rows x columns, values in the right cells), empty-chunk rate, header/footer leakage. Golden documents with known probe facts are asserted on every parser change.

1. **Retrieval quality.** Does the system find the right content?

Measured as hit rate and mean reciprocal rank over a golden Q&A set built from our own corpus, sliced by document and query type.

1. **Answer quality.** Is the response correct and honest? Expected-fact

checks for extractive answers; correctness, faithfulness, and abstention checks for generated answers. Unanswerable questions must refuse, not invent.

1. **Production monitoring.** Every query logged; abstention rate,

zero-result rate, user downvote rate, and latency on a dashboard, with nightly re-runs of the golden set alerting on regression.

The full methodology, including the golden-set construction protocol and sample-size guidance, is documented alongside the pilot code.

What is already proven

I built a pilot pipeline this week: documents parsed through a production document API, normalized into a searchable knowledge base, with a web Q&A interface. Against a real 10-Q filing, the baseline measures retrieval hit rate 1.00 with MRR 0.93, answer score 0.78, and clean parse health (75 chunks, zero empty, 26 table-bearing). The framework has already caught and fixed two real defects, including a case where the system answered an unanswerable question instead of abstaining.

Cost

Parsing 10,000 documents (roughly 10 pages each) costs on the order of $1,200 using a mixed accuracy/cost engine profile, plus roughly $50 per month in infrastructure. The evaluation buildout itself is primarily human time: about a day to establish the first golden set, then minutes per week to maintain. Compared against the cost of acting on wrong numbers, it is the cheapest insurance in the system.

Ask

1. Approve this methodology as the starting standard for document AI

quality on our team.

1. Next steps on approval: grow the golden set to decision-grade size

(~50 cases), add nightly regression runs, and extend query logging to production traffic so the eval set grows out of real usage.